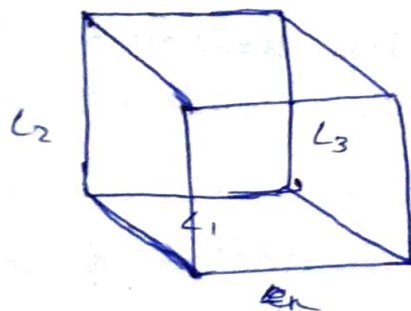


20-1-25
→ wire frame modelling

→ solid modelling $\left\{ \begin{array}{l} \text{Constructive solid geometry (CSG or C-Rep)} \\ \text{Boundary representation method. [B-Rep].} \end{array} \right.$

→ surface modelling



⇒ wire frame

adv:

→ Least computational capacity.

disadv:

→ Confusion of faces (front, top) or difficult to identify faces.

→ cannot calculate mass properties.

→ Both topological & geometrical data need to be given.

Geometric model.

The user or designer construct a geometric model of an object on a CAD system by inputting the object data as required by the modelling technique via user interface (GUI) provided by the software. The software convert such data into mathematical representation which store in the model database.

wireframe model.

The image assumes appearance of a frame constructed out of wire hence named as wireframe.

model.

- simplest model. that can be represented mathematically
- of an object.
- It is an edge representation of an object.
- Consists of points, lines, arcs, circles, conics & curves.
- simplicity to construct, doesn't require computational time and modeling
- Considered as a natural extension of traditional method of drafting. - doesn't require extensive training.
- Disadv:
 - Usually ambiguous representation of real object.
 - very difficult for human interpretation.
 - Models of complex designs having many edges become very confusing & difficult to interpret. This can be avoided to a certain extent by introduction of automatic hidden line removal feature.
 - Lack of visual coherence & info to determine object profile.
 - Volume & mass properties cannot be calculated.
 - Both topological & geometrical data are needed to construct wireframe model.

Note: Topology info contain connectivity & associativity of object entities. (relations like $L_1 \perp^{45}$ to L_2)

Geometric info contains actual dimensions of the object.

What we do in CAD is generally Boundary representation.

For CSG, primitives will be there in the modelling software's library.

Solid modelling

→ More complete representation than surface & wireframe model.

→ Permit fractional automation & integration.

Eg - mass property calculation is possible.

→ Solid model can be quickly created without having to define individual locations like in the case of wire frame model.

22-1-25 → Solid model contain 2 basic approach (schemes)

1. Constructive Solid Geometry (CSG) (C-rep). also called building block approach.

2. Boundary representation (B-rep)

The CSG system allows the user to build the model out of solid graphic primitives such as rectangular blocks, cubes, spheres, cylinders, pyramids etc. The boundary representation approach requires the user to draw the outline or boundary of the object.

The designer have to sketch various views of the object, drawing inter connecting lines among the views, various transformation to get the desired shape
apply

CAD

Rojin Sir

22-1-25
The C-rep system ^{is} relatively easy to construct by adding, subtracting & intersecting the components. The B-rep system ^{is} easy to model complex shapes. Comparatively require more storage but less computational effort.

Different types of modelling software.

Autodesk - Fusion 360, Tinker CAD, Auto CAD.

Dassault system - Catia, Solid works.

Siemens. - Siemens NX, Solid edge

Free CAD, Opens CAD. → free wares.

Engineering analysis

In the formulation of nearly any engg design project, some type of analysis is required. The analysis may involve stress-strain calculation, heat transfer computation, kinematic mechanism synthesis etc.

The comp. can be used to ~~aim~~^{aim} in the analysis work. It is often necessary that specific prgram being developed internally by engg analysis grp to solve particular design problems.

29-1-25.

CAD/CAM Database.

- CAD system offers opportunity to develop database needed to manufacture the product.
- An integrated CAD/CAM Database systems is a direct link established b/w product design & manufacturing.
- Geometric data exchange refers to conversions from one geometric data format to another.
- Fundamental incompatibilities among entity representation further complicate exchanging the modelling data.

Need for geometric data exchange

- Heterogeneous expertise in industry.
People may use different CAD package as long as alternatives are available $\therefore$ data exchange is needed.
- Use of application specific packages.
data need to be exchanged from 1 module to another.

- Migration from 1 system to another.
- Data exchange with collaborators / customers / suppliers.

Options available for data exchange

- Native format.
- Neutral "
- Standard "
- Binary "
- ASCII format.

Std available formats

IGES ~~std~~ - Initial Graphics Exchange Format

DXF - Drawing exchange format.

STL - standard triangular language.

STEP - standard for Exchange of Products data.

PDES, ACIS, Parasolid.

- IGES → 1st std exchange format.

Each version must remain upward compatible with previous version.

Translate native database to IGES format - preprocessor.

{ opposite - Post processor.

→ Translators.

exchanged data

Design data - info that designers generate from geometric models. Mass property, Finite element mesh data, ...

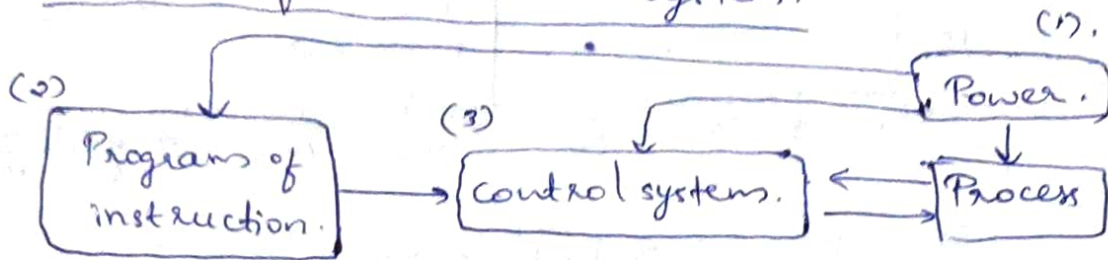
Manufacturing data - info as tooling, NC tool path.

Tolerancing, process planning, tool
designs, bill of materials...

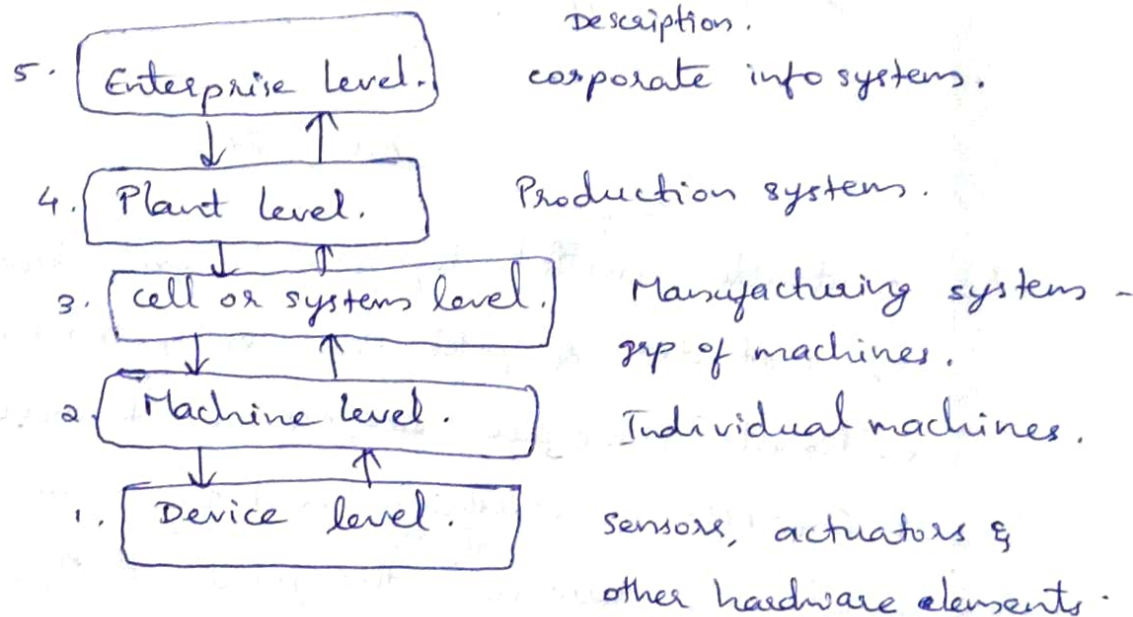
5-1-25.

Automation.

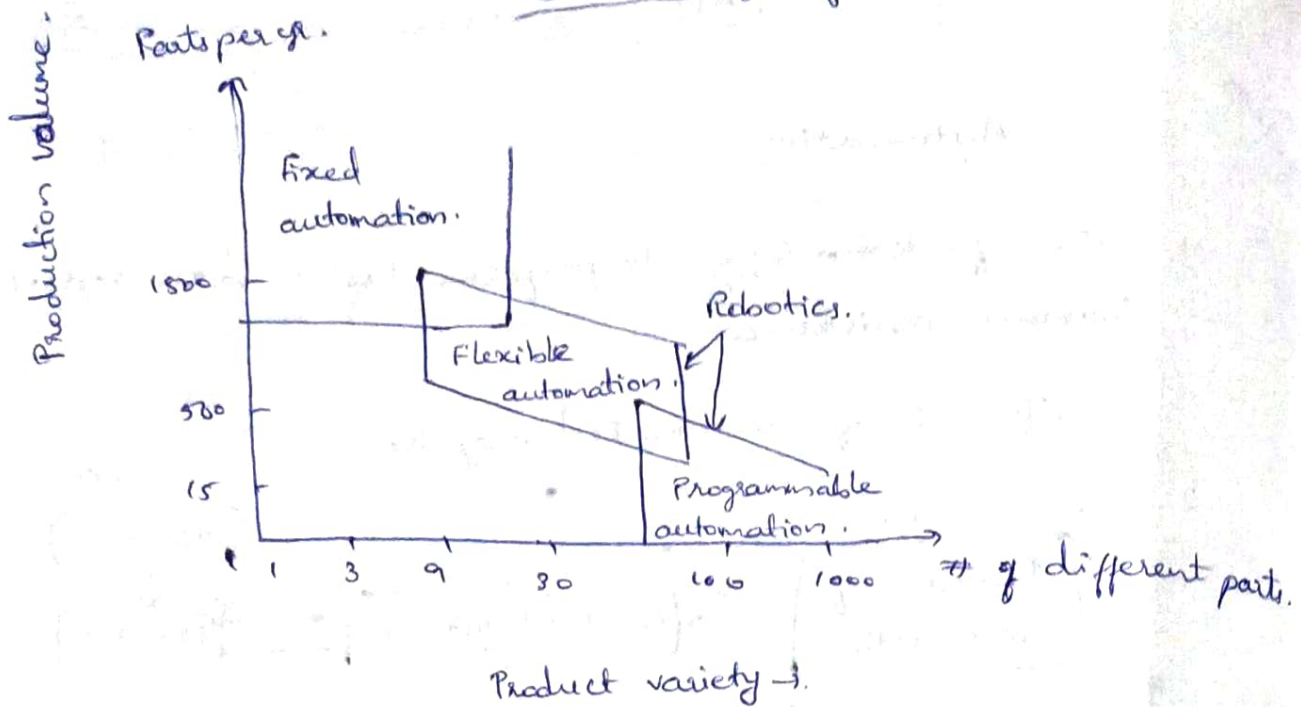
- Use of technology to perform tasks with reduced human assistance.

Elements of automated system.Levels of automation.

- Based on complexity of systems being automated.



3 basic types of automation.



In fixed automation, single part is manufactured in large quantity.

Automated flow lines.

- Consists of several machines / workstations which are linked together by work handling devices that transfer parts b/w these stations.
- Raw work-part enters (end of the line & processing steps are performed sequentially as part moves from 1 station to other.
- It is possible to incorporate buffer storage zones into flow line either at single location / b/w every w.s.
- Also possible to include inspection stations in line to automatically perform intermediate checks on quality.

1. In-line type.

- Consists of sequence of w.s. more or less straight line arrangement.
- Ideal for high volume production.

2. Segmented in-line type.

- Consists of 2 or more straight line arrangement usually J or to each other with $\angle$ or U or  shape.
- More flexibility
- Flow of work can take a few 90° turns either for workpiece reorientation, factory layout limitation...

3

Rotary type

- workparts are indexed around a circular table/dial
- vs. stationary.



Transfer Mechanisms.

- To move.

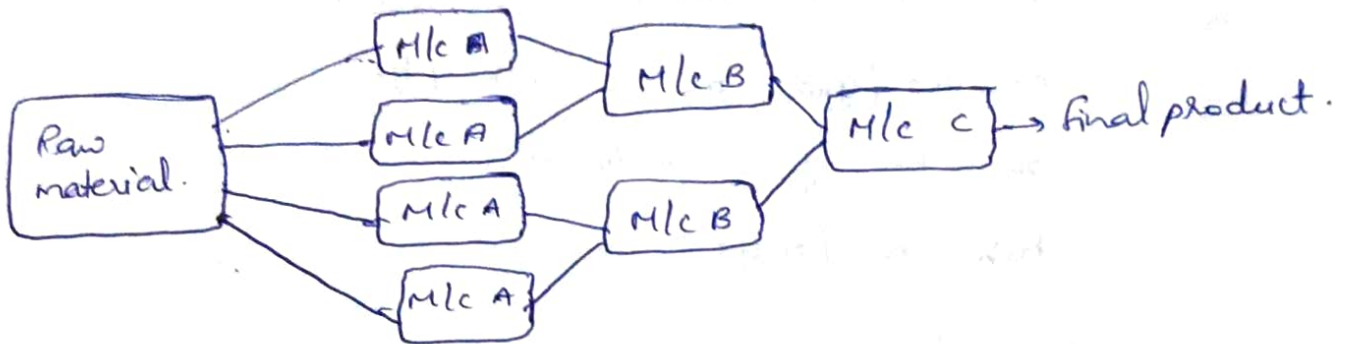
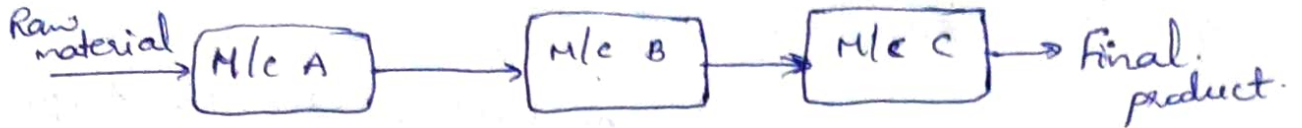
Linear travel for in line machine.

Rotary motion for dial indexing machines.

- a). Walking beam system.
 - b). Powered roller conveyor system.
 - a). Rotary transfer.
 - b). Ratchet & Pawl.
 - c). Geneva mechanism.
 - d). Cam.
- } $R \rightarrow L$
- } $R \rightarrow R$
- $R \rightarrow R$

CAD

Rojin Sir
CAD & CAM by
Michael Greaves



$$M/c A = 25 \text{ prod/hr.}$$

$$M/c B = 50 \text{ prod/hr.}$$

$$M/c C = 100 \text{ prod/hr.}$$

Line balancing

Main objective of line balancing is to distribute the task evenly over workstation so that ideal time of man & machine is minimized. The line balancing aims at grouping the facilities & workers in an efficient pattern in order to obtain an optimum balance of the capacity.

2 imp concept of line balancing are:

- 1). Separation of total work content into min rational work element.
- 2). Precedence constraints must satisfy.

$$T_{wc} = \sum T_{ek}$$

$$T_{wc} = \text{work content time.}$$

ΣT_{ek} = Sum of work content time.

$$Q_p = \frac{D_a}{50 S_w H_{sh}}$$

R_p = avg. hourly productive rate (units/year).

D_a = Annual demand (units/year).

50 wks/year or 52 wks/yr.

S_w = no. of shifts/week.

H_{sh} = Hr/shift.

$$T_c = \frac{60E}{R_p}$$

T_c = cycle time (min/cycle).

E = uptime efficiency for line.

R_p = Production rate.

$$W^* = \text{min no of workers} = \frac{T_{wc}}{T_c}$$

$$E_b = \frac{T_{wc}}{W T_s}$$

E_b = Balance efficiency.

W = Actual no. of workers.

T_s = station or service time.

$$d = \frac{W T_s - T_{wc}}{W T_s}$$

d = balance delay.

Note:

A perfect line balance, the balance efficiency,

$$E_b = 100\% \text{ and } E_b + d = 1.$$

$$1. \quad T_{wc} = \sum T_{ek} = 4 \text{ min.} \quad (Pg-4)$$

$$\text{Then } R_p = \frac{D_a}{50 \times S_w \times H_{sh}} = \frac{D_a}{50 \times 5 \times 7.5} = \underline{\underline{53.33}}$$

$$T_c = \frac{60E}{P}$$

Groover
automation.

$$W^* = \frac{T_{wc}}{T_c} = \frac{4}{1.08} = 3.7 \approx \underline{\underline{4}}$$

$$T_s = T_c - T_a = 1.08 - 0.08 = \underline{\underline{1 \text{ min.}}}$$